

**SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ**

COURSE NAME: COMPUTER VISION

COURSE CODE:ITA0515

Slot B

COURSE OUTCOME COVERED

CO3: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)

Assessment Tool Weightage: 10%

QUESTIONS:50

Total Marks 50

Duration :60 Minutes

Annexure A

MCQ

Code of Conduct:

I M.Pranay (192525382)_certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:M.Pranay

Rubrics for Evaluation

Criteria	Weightae (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect	
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers	
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts	
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time	
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions	

1. Image Normalization (Preprocessing)

An image pixel value $r = 150$ is normalized to the range $[0,1]$ from 8-bit data.

What is the normalized value?

- A) 0.39
 - B) 0.49
 - C) 0.59
 - D) 0.78
- A. 0.59 (C)

2. Image Storage Calculation

A grayscale image of size **1024 × 512** is stored using **8 bits per pixel**.

What is the storage requirement?

- A) 256 KB
 - B) 512 KB
 - C) 1024 KB
 - D) 2048 KB
- A. 512 KB (B)

3. Edge Detection – Gradient Magnitude

Given Sobel gradients:

($G_x = 6$), ($G_y = 8$)

What is the gradient magnitude?

- A) 10
 - B) 12
 - C) 14
 - D) 16
- A.10 (A)

4. Edge Direction Calculation

For the same gradients ($G_x = 6$), ($G_y = 8$),
what is the **edge direction**?

- A) 36.8°
 - B) 45°
 - C) 53.1°
- A. 53.1° (C)

5. Harris Corner Detection Response

Given:

$$R = \det(M) - k(\text{trace}(M))^2$$

Where:

$$\det(M) = 200, \text{trace}(M) = 30, k = 0.04$$

What is the corner response R?

- A) 136
 - B) 164
 - C) 180
 - D) 200
- A.164 (B)

6. Harris Corner Classification

If the Harris response $R = +250$, the pixel is classified as:

- A) Flat region
 - B) Edge
 - C) Corner
 - D) Noise
- A. Corner (C)

7. Hough Transform – Line Representation

A line detected at $\theta = 90^\circ$ and $\rho = 50$ corresponds to:

- A) Horizontal line at $y = 50$
 - B) Vertical line at $x = 50$
 - C) Diagonal line
 - D) Random line
- A. Horizontal line at $y = 50$ (A)

8. An image is scaled by a factor of 2.

If original keypoint scale is $\sigma = 1.6$, new scale is:

- A) 0.8
 - B) 1.6
 - C) 3.2
 - D) 4.0
- A. 3.2 (C)

9. PCA Dimensionality Reduction

A feature vector has **100 dimensions**.

PCA retains **95% variance using 20 components**.

What is the dimensionality after PCA?

- A) 5
 - B) 20
 - C) 80
 - D) 95
- A. 20 (B)

10. PCA Covariance Matrix Size

For a dataset with **50 features**, what is the size of the covariance matrix?

- A) 50×1
 - B) 1×50
 - C) 50×50
 - D) 100×100
- A. 50×50 (C)

11. Image Quantization (Preprocessing)

A grayscale image is quantized from **256 levels to 32 levels**.

What is the **number of bits per pixel** after quantization?

- A) 3
 - B) 4
 - C) 5
 - D) 6
- A. 5 bits (C)

12. Image Storage Reduction

An RGB image of size **512 × 512** uses **8 bits per channel**.
It is converted to grayscale with **8 bits per pixel**.

What is the **percentage reduction in storage**?

- A) 33%
 - B) 50%
 - C) 66.7%
 - D) 75%
- A. 66.7% (C)

13. Sobel Edge Response

Given Sobel kernel response values:
($G_x = -12$), ($G_y = 5$)

What is the **gradient magnitude (approx.)**?

- A) 10
 - B) 12
 - C) 13
 - D) 15
- A.13 (C)

14. Gradient Direction Calculation

For ($G_x = -12$), ($G_y = 5$), the gradient direction is closest to:

- A) 22°
 - B) 157°
 - C) 180°
 - D) -22°
- A. 157° (B)

15. Harris Corner Detection Response

Given:

$$\det(M) = 400$$

$$\text{trace}(M) = 40$$

$$k = 0.06$$

Compute the Harris response **R**.

- A) 256
 - B) 304
 - C) 352
 - D) 400
- A.304 (B)

16. Harris Corner Decision

For a pixel, eigenvalues of matrix **M** are:

$$\lambda_1 = 120, \lambda_2 = 130$$

This pixel is classified as:

- A) Flat region
 - B) Edge
 - C) Corner
 - D) Noise
- A. Corner (C)

17. Hough Transform Voting

A point contributes votes to **180 values of θ** in Hough space.

What is the **shape of its representation**?

- A) A point
 - B) A straight line
 - C) A sinusoidal curve
 - D) A circle
- A. A sinusoidal curve (C)

18. Hough Line Detection

A detected line has parameters **$p = 0, \theta = 45^\circ$** .

Which line is represented?

- A) Horizontal through origin
 - B) Vertical through origin
 - C) Diagonal line through origin
 - D) Random orientation
- A. Diagonal line through origin (C)

19. SIFT Keypoint Descriptor Size

Each SIFT descriptor consists of **4×4 histograms**, each with **8 bins**.

What is the descriptor dimensionality?

- A) 64
 - B) 96
 - C) 128
 - D) 256
- A.128 (C)

20. PCA Eigenvalue Selection

Eigenvalues of covariance matrix:

[40, 25, 15, 10, 5, 5]

How many principal components are needed to retain **80% variance**?

A) 2

B) 3

C) 4

D) 5

A.3 (B)

21. Preprocessing – Normalization Scenario

A surveillance camera outputs **8-bit grayscale images**.

Before feature extraction, pixel values must be normalized to **[0,1]**.

A pixel has value **204**.

What is the normalized value?

A) 0.60

B) 0.70

C) 0.80

D) 0.90

A. 0.80 (C)

22. Image Representation Scenario

A medical image of size **2048 × 1024** is stored using **16 bits per pixel**.

What is the **memory required**?

A) 2 MB

B) 4 MB

C) 8 MB

D) 16 MB

A.4 MB (B)

23. Edge Detection Scenario

At a pixel location, Sobel gradients are:

($G_x = 9$), ($G_y = 12$)

What is the **edge magnitude** at that pixel?

A) 15

B) 18

C) 21

D) 25

A. 15 (A)

24. Edge Orientation Scenario

Using the same gradients ($G_x = 9$), ($G_y = 12$),

what is the **edge direction (approx.)**?

- A) 36.8°
- B) 45°
- C) 53.1°
- D) 60° A. 53.1° (C)

25. Harris Corner Detection Scenario

For a region in an image, the Harris matrix values are:

$\det(M) = 300$, $\text{trace}(M) = 25$, $k = 0.04$

What is the **Harris response R**?

- A) 200
- B) 250
- C) 275
- D) 300 A. 275 (C)

26. Harris Corner Interpretation Scenario

If the Harris response **R is large and positive**, the pixel corresponds to:

- A) Flat region
- B) Edge
- C) Corner
- D) Noise A. Corner (C)

27. Hough Transform Scenario

A point in image space votes for all possible **θ values** in Hough space.

What shape does this vote form?

- A) A point
- B) A straight line
- C) A sinusoidal curve
- D) A circle A. A sinusoidal curve (C)

28. Hough Line Detection Scenario

A detected line has parameters **$p = 40$, $\theta = 90^\circ$** .

Which line is represented?

- A) Vertical line $x = 40$
- B) Horizontal line $y = 40$
- C) Diagonal line A. Horizontal line $y = 40$ (B)

29. SIFT Feature Scenario

An image is scaled by a factor of **0.5**.

A SIFT keypoint detected at scale $\sigma = 2.0$ originally will now appear at:

- B) 0.5
 - A) 1.0
 - C) 2.0
 - D) 4.0
- A.1.0 (A)

30. PCA Dimensionality Reduction Scenario

A feature vector has **200 dimensions**.

After PCA, **30 principal components** retain **92% variance**.

What is the **new feature dimension**?

- A) 30
 - B) 92
 - C) 170
 - D) 200
- A.30 (A)

31. Preprocessing – Bit Depth Conversion

A grayscale image originally stored in **12-bit format** is converted to **8-bit** before processing.

An original pixel value is **3000**.

What is the approximate 8-bit value after linear scaling?

- A) 120
 - B) 150
 - C) 187
 - D) 225
- A.187 (C)

32. Image Representation – Sampling Scenario

A camera samples an image at **600 × 400** pixels.

The image is down-sampled by a factor of **2 in both directions**.

What is the new image size?

- A) 150 × 100
 - B) 300 × 200
 - C) 400 × 300
 - D) 600 × 400
- A. 300 × 200 (B)

33. Edge Detection – Sobel Response

At a pixel, Sobel gradients are:

($G_x = -8$), ($G_y = 15$)

What is the **edge magnitude (approx.)**?

- A) 13
 - B) 15
 - C) 17
 - D) 20
- A. 17 (C)

34. Edge Orientation Scenario

For ($G_x = -8$), ($G_y = 15$), the gradient direction is closest to:

- A) 28°
 - B) 62°
 - C) 118°
 - D) 152°
- A. 118° (C)

35. Harris Corner Detection – Computation

Given:

$\det(M) = 500$, $\text{trace}(M) = 35$, $k = 0.05$

What is the Harris response **R**?

- A) 439
 - B) 450
 - C) 469
- A. 439 (C)

36. Harris Corner Classification

If one eigenvalue is large and the other is small, the pixel represents:

- A) Flat region
 - B) Edge
 - C) Corner
 - D) Noise
- A. Edge (B)

37. Hough Transform – Line Interpretation

A line detected with parameters $\rho = 0$, $\theta = 90^\circ$ corresponds to:

- A) Vertical line at $x = 0$
 - B) Horizontal line at $y = 0$
 - C) Diagonal line
- A. Vertical line at $x = 0$ (A)

38. SIFT Descriptor Scenario

Each SIFT descriptor contains 4×4 blocks with **8 orientation bins** each.

How many values describe one keypoint?

- A) 32
 - B) 64
 - C) 96
 - D) 128
- A.128 (D)

39. PCA Variance Retention

Eigenvalues of covariance matrix are:

[50,; 30,; 10,; 5,; 5]

How many principal components are needed to retain **90% variance**?

- A) 1
 - B) 2
 - C) 3
 - D) 4
- A.3 (C)

40. PCA Projection Scenario

Original feature vector dimension = **120**.

PCA reduces it to **25 components**.

What is the **reduction percentage**?

- A) 60%
 - B) 70%
 - C) 79%
- A. 79% (C)

41. Quantization Error Analysis (Preprocessing)

An 8-bit grayscale image is quantized to **64 levels**.

Original pixel value = **150**.

What is the **maximum possible quantization error**?

- A) ± 1
 - B) ± 2
 - C) ± 4
 - D) ± 8
- A. ± 2 B()

42. Image Representation Storage Comparison

Image A: **1024 × 1024 grayscale (8-bit)**

Image B: **1024 × 1024 RGB (8-bit per channel)**

What is the **storage ratio (B : A)**?

- A) 1 : 1
- B) 2 : 1
- C) 3 : 1
- D) 4 : 1 A. 3 : 1 (C)

43. Sobel Edge Strength Comparison

At two pixels:

Pixel-1: ($G_x = 6$,; $G_y = 8$)

Pixel-2: ($G_x = 9$,; $G_y = 12$)

Which pixel corresponds to a **stronger edge**?

- A) Pixel-1
- B) Pixel-2
- C) Both equal A. Pixel-2 (B)

44. Harris Corner Response Comparison

Two regions have the following Harris parameters:

Region **det(M)** **trace(M)**

R1 300 25

R2 350 40

Given $k = 0.04$, which region is **more likely a corner**?

- A) R1
- B) R2
- C) Both
- D) Neither A. R2 (B)

45. Harris Corner vs Edge Decision

Eigenvalues of matrix **M** at a pixel are:

$\lambda_1 = 120$, $\lambda_2 = 10$

What does this pixel represent?

- A) Flat region
- B) Edge
- C) Corner
- D) Noise A. Edge (B)

46. Hough Transform Voting Analysis

Two points generate curves in Hough space that intersect at ($\rho = 60$, $\theta = 45^\circ$).

What does this intersection indicate?

- A) A corner
 - B) A circle
 - C) A line in image space
 - D) Noise accumulation
- A. A line in image space (C)

47. Hough Transform Parameter Effect

Increasing the θ resolution from 1° to 0.5° results in:

- A) Fewer accumulator cells
 - B) Lower memory usage
 - C) Higher line detection accuracy
- A. Higher line detection accuracy (C)

48. SIFT Scale Analysis

An object appears at scale $\sigma = 1.6$ in Image-1.

In Image-2, the object is scaled by $1.5\times$.

At what scale should the corresponding SIFT keypoint appear?

- A) 1.1
 - B) 1.6
 - C) 2.4
- A. 2.4 (C)

49. PCA Variance Contribution Analysis

Eigenvalues of covariance matrix are:

[60,; 20,; 10,; 5,; 5]

How many principal components are needed to retain **90% variance**?

- A) 1
 - B) 2
 - C) 3
 - D) 4
- A. 3 (C)

50. PCA Dimensionality Reduction Impact

Original feature dimension = **150**

After PCA = **30 components**

What is the **percentage reduction**?

- A) 60%
- B) 70%

C) 80%

D) 85%

A. 80% (C)